

Date: 12/09/2026

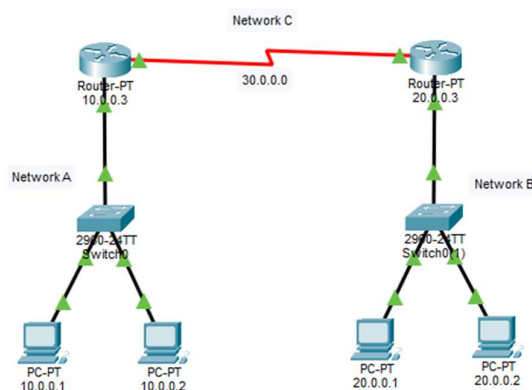
Lab Practical #12:

Study the concept of routing using packet tracer. **(Dynamic Routing)**

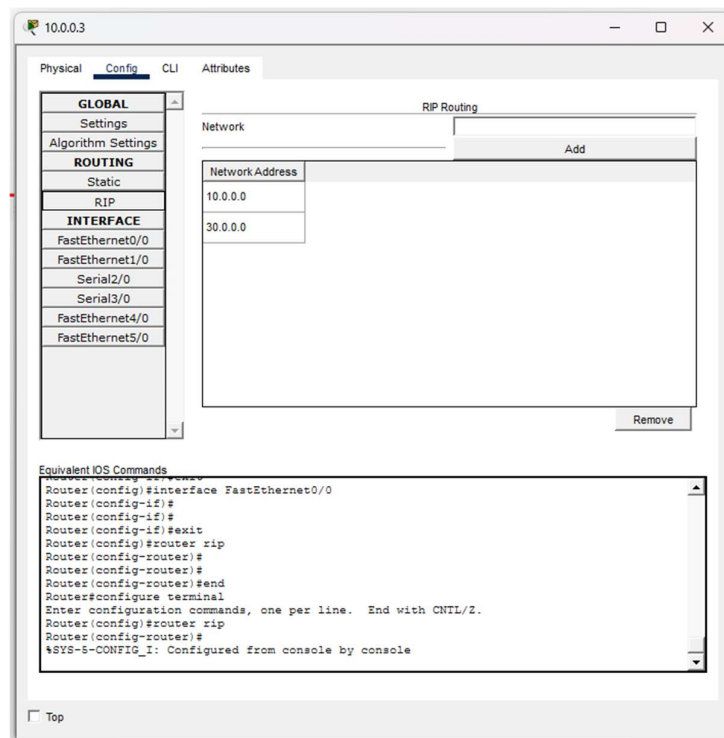
Practical Assignment #12:

A. Connect the **two different routers** based on the calculated IP addresses and subnet using a packet tracer.

1. Topology Screenshot



2. RIP routing screenshot with routing table.



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3. IP address table of each node and network ID of each network.

Node	IP address	Network	Network ID
PC0	10.0.0.1	Network A	10.0.0.0
PC1	10.0.0.2	Network B	20.0.0.0
PC3	20.0.0.1	Network C	30.0.0.0
PC4	20.0.0.2		
Router0(Fa0/0)	10.0.0.3		
Router0(Se2/0)	30.0.0.1		
Router1(Fa0/0)	20.0.0.3		
Router1(Se2/0)	30.0.0.2		

4. Packet (PDU) transfer screenshot

PDU Information at Device: 10.0.0.2

OSI Model [Inbound PDU Details](#)

PDU Formats

EthernetII

0 4 8 Bytes

PREAMBLE: 101010...10 DEST ADDR: 000D.BD6C.37C3

SRC ADDR: 000B.BEB3.A5B TYP E:0 DATA (VARIABLE LENGTH) FCS: 0x00000000

IP

0 4 8 16 20 24 Bits

VER: 4 IHL: 5 DSCP: 0x00 TL: 28

ID: 0x0008 FLA FRAG OFFSET: 0x000

TTL: 126 PRO: 0x01 CHKSUM

SRC IP: 20.0.0.1

DST IP: 10.0.0.2

DATA (VARIABLE LENGTH)

ICMP

0 8 16 Bits

TYPE: 0x00 CODE: 0x00 CHECKSUM

ID: 0x0004 SEQ NUMBER: 3

Variable Size PDU

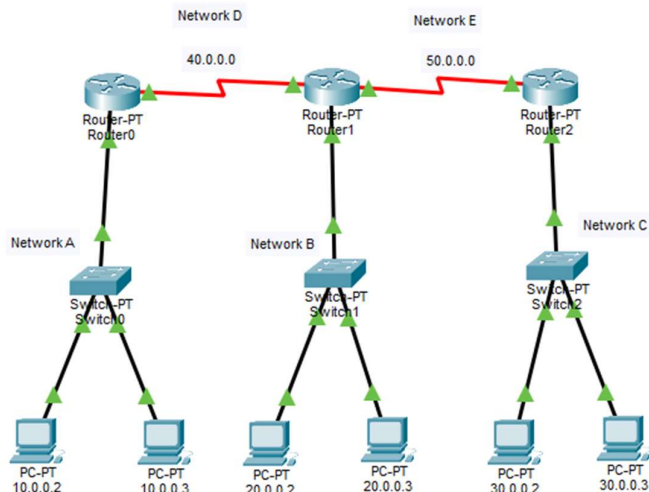
0 8 16 Bytes

DATA (VARIABLE LENGTH)

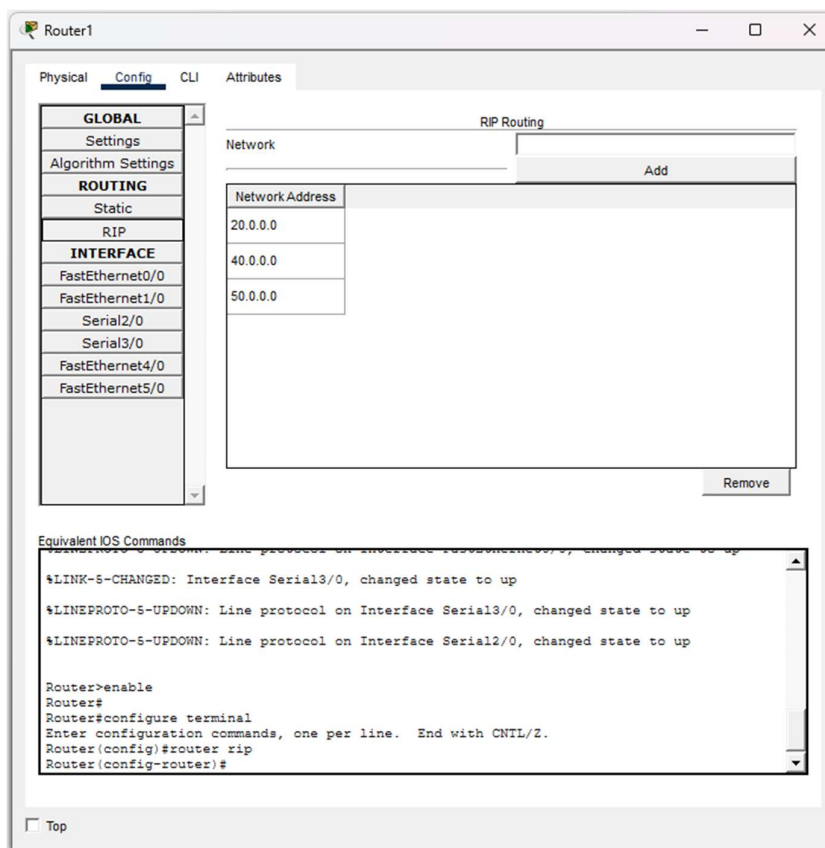
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B. Connect the **three different routers** based on the calculated IP addresses and subnet using a packet tracer.

1. Topology Screenshot



2. RIP routing screenshot with routing table.

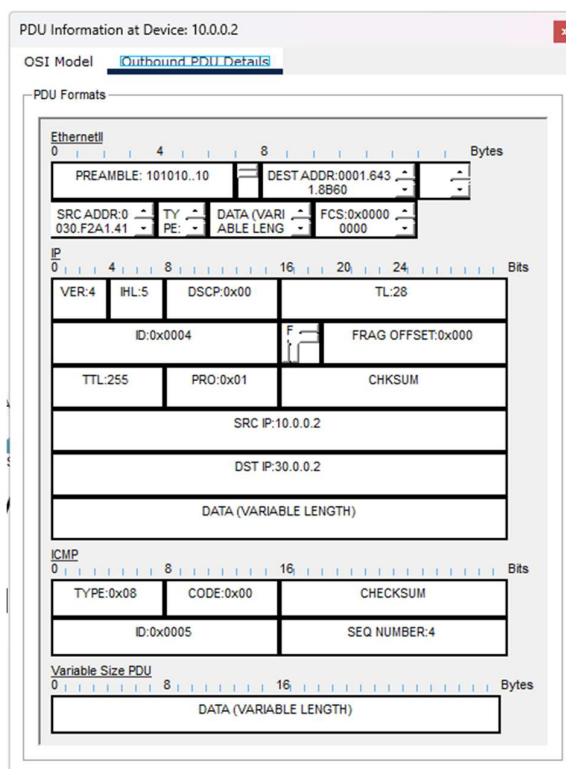


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3. IP address table of each node and network ID of each network.

Node	IP address	Network	Network ID
PC0	10.0.0.2	Network A	10.0.0.0
PC1	10.0.0.3	Network B	20.0.0.0
PC3	20.0.0.2	Network C	30.0.0.0
PC4	20.0.0.3	Network D	40.0.0.0
PC5	30.0.0.2	Network E	50.0.0.0
PC6	30.0.0.3		
Router0(Fa0/0)	10.0.0.1		
Router0(Se2/0)	40.0.0.1		
Router1(Fa0/0)	20.0.0.1		
Router1(Se2/0)	40.0.0.2		
Router1(Se3/0)	50.0.0.1		
Router2(Fa0/0)	30.0.0.1		
Router2(Se2/0)	50.0.0.2		

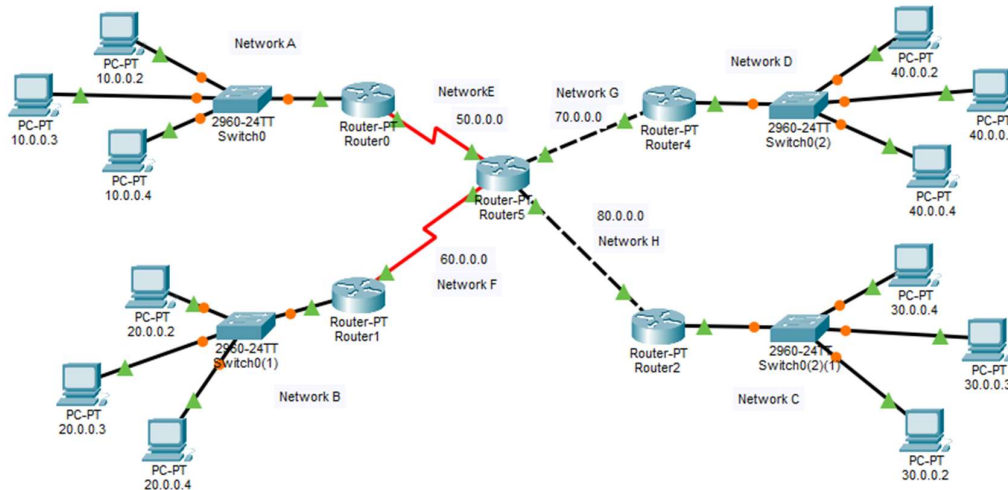
4. (PDU) transfer screenshot



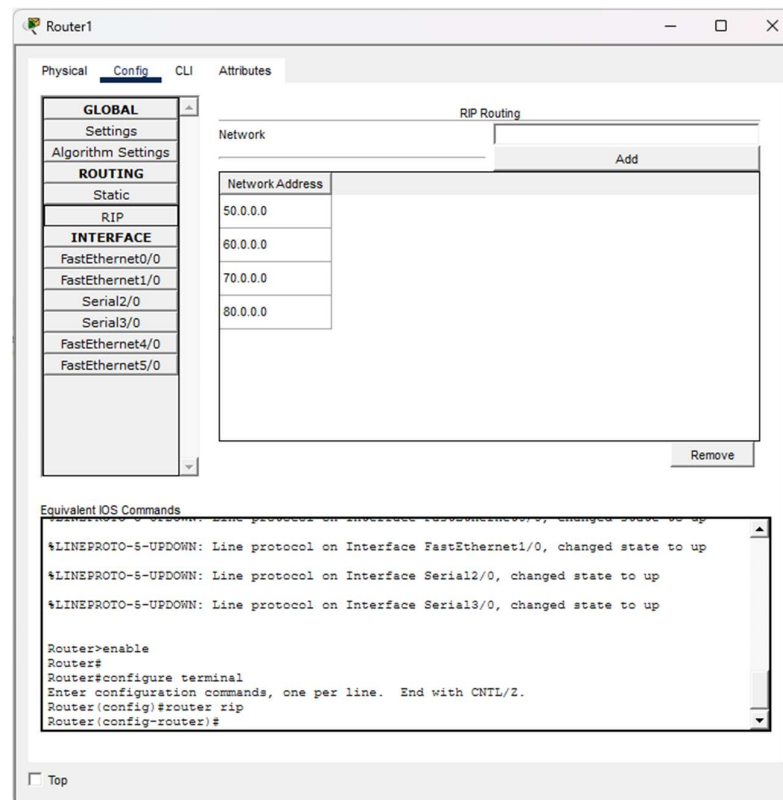
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C. Connect the **five different routers** based on the calculated IP addresses and subnet using a packet tracer.

1. Topology Screenshot



2. RIP routing screenshot with routing table.





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3. IP address table of each node and network ID of each network.

Node	IP address	Network	Network ID
PC0	10.0.0.2	Network A	10.0.0.0
PC1	10.0.0.3	Network B	20.0.0.0
PC3	10.0.0.4	Network C	30.0.0.0
PC4	20.0.0.2	Network D	40.0.0.0
PC5	20.0.0.3	Network E	50.0.0.0
PC6	20.0.0.4	NetworkF	60.0.0.0
PC7	30.0.0.2	NetworkG	70.0.0.0
PC8	30.0.0.3	NetworkH	80.0.0.0
PC9	30.0.0.4		
Router0(Fa0/0)	10.0.0.1		
Router0(Se2/0)	50.0.0.1		
Router1(Fa0/0)	20.0.0.1		
Router1(Se2/0)	60.0.0.1		
Router2(Fa0/0)	30.0.0.1		
Router2(Fa1/0)	70.0.0.1		
Router4(Fa0/0)	40.0.0.1		
Router4(Fa1/0)	80.0.0.1		
Router5(Fa0/0)	70.0.0.2		
Router5(Fa1/0)	80.0.0.2		
Router5(Se2/0)	60.0.0.2		
Router5(Se3/0)	50.0.0.2		

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4. (PDU) transfer screenshot.

